Keivalya Pandya

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Education

Northeastern University **Dec 2026**
Master of Science, Robotics (Computer Science)
- **GPA:** 3.952
- **Coursework:** Deep Learning, Robot Mechanics and Control, Mechanistic Interpretability, Reinforcement Learning, Mobile Robotics, Master's Research (Imitation Learning)

Birla Vishvakarma Mahavidyalaya **Jun 2023**
Bachelor of Technology, Mechanical Engineering
- **Achievements:** Secured State Government Scholarship Tuition Fee Waiver
- **Coursework:** Mechatronics, Control Systems, Applied Electronics, CAD/CAM, Material Science, Deep Learning

Experience

General Motors
Robotics Engineering Intern *Mountain View, CA* · **May 2026 - Aug 2026**
- Develop VLA and WAM control algorithms for a bi-manual manufacturing robot resulting in more adaptable manufacturing processes
- Lead data collection, pre-processing and data ablation studies on UR20, controlled using YAM arm.
- Benchmarking 7 open source models in tricky and unpredictable environments (in both, hardware + software)

Familiar Machines & Magic
Reinforcement Learning Intern *Boston, MA* · **Sep 2025 - Dec 2025**
- Designed, implemented, and tested learning algorithms using IsaacLab and sim2real techniques to enhance autonomous robotic behaviors, resulting in improved performance in real-world environments
- Worked with AI, Hardware, and Software teams to develop embedded firmware, unit tests, and simulation scripts, enhancing policy optimization and improving robot efficiency

Northeastern University
Robotics AI Researcher **May 2025 - Aug 2025**
- Developed a human-in-the-loop reinforcement learning framework (RLHF) in NVIDIA Isaac Sim, using Python, which enabled the mobile robot to join ongoing human-robot interactions safely and improved overall interaction safety
- Proposed and implemented a pose generation pipeline that applied spatial heuristics (distance, area, etc) and MLP-based scoring, resulting in more accurate pose selection and faster planning for robot perception tasks

Research Apprentice **Jan 2025 - Apr 2025**
- Deployed dexterous manipulation policies on Hello-Robot Stretch3, benchmarking Diffusion Policy, VQ-BeT, and ACT to enhance human-robot interaction capabilities
- Analyzed Vision-Language-Action models and implemented OpenVLA and RUM on Stretch3, contributing to more robust model deployment and testing processes
- Selected as one of 40 among 20,000+ students for fully funded research at Khoury College of Computer Science

BVM Engineering
Robotics Engineer, Full-Stack Developer **Dec 2023 - Aug 2024**
- Created an E-commerce platform for institutional component sales and inventory management, improving transaction processing speed and stock accuracy with features like automated alerts and analytics
- Developed modular centralized control software with OPC-UA protocol for static stations and AMRs, optimizing unmanned assembly lines and eliminating human interference and delays

Projects

mini-VLA
- Built an end-to-end Vision-Language-Action policy that learns instruction-conditioned robot manipulation from demonstrations using diffusion-based action generation.
- Integrated vision, language, and proprioceptive encoders into a lightweight (~150 LOC core), modular framework enabling fast iteration and research prototyping in simulation.

Determining Language Grounding Failures in Vision-Language-Action Models
- Developed a scene-fixed causal evaluation protocol to isolate language dependencies across VLA control policies up to 7.5B parameters.
- Identified critical control policy vulnerabilities using conditional task entropy, quantifying a 31.7 percentage point performance drop under contrastive action perturbations.

RoboSpace.app
- Simulate and train any robot simply by dragging and dropping its XML/URDF model onto the browser (no installation required).
- Gained 1000+ unique users and acquired 200+ registered users within 36 hours of launching the app.

Imitation Learning Robot Arm Manipulation using BC
- Developed 3-phase SAC for long-horizon, continuous-action MuJoCo environment for 9-DoF Franka.
- Robot learns cooking-related tasks, trained on human demonstrations using Behavior Cloning (BC).

Saksham – Prosthetic limb with remote sensors
- IoT-enabled data and performance monitoring, built with tactile sensors for sense of touch, dedicated web-app and versatile grip.
- Patent-granted technology (design and utility) funded by Government of India.

SE(3) Inverse Kinematics for 7-DOF Franka Panda
- Built a custom SE(3) Newton Inverse Kinematics solver and analytical Forward Kinematics model in MuJoCo for 7-DOF robotic control.
- Achieved sub-micrometer tracking on complex helical trajectories with 1.1 x 10^-4 mm position and 2.5 x 10^-4 degree orientation steady-state errors.

Skills

- **Skills**: Python, C++, ROS2, Pytorch, Tensorflow, OpenCV, VLA, Reinforcement Learning, Perception, Deep Learning, Machine Learning, MuJoCo, SLAM, Imitation Learning, Gazebo, Nvidia Isaac Sim, MATLAB, Simulink, Docker, FastAPI, VLMs

Patent

- Pandya K. **IN Patent** 356336-001/125869. AI-based Bio-electro-mechanical Prosthetic Limb.